

Harness CI/CD AI Customizations

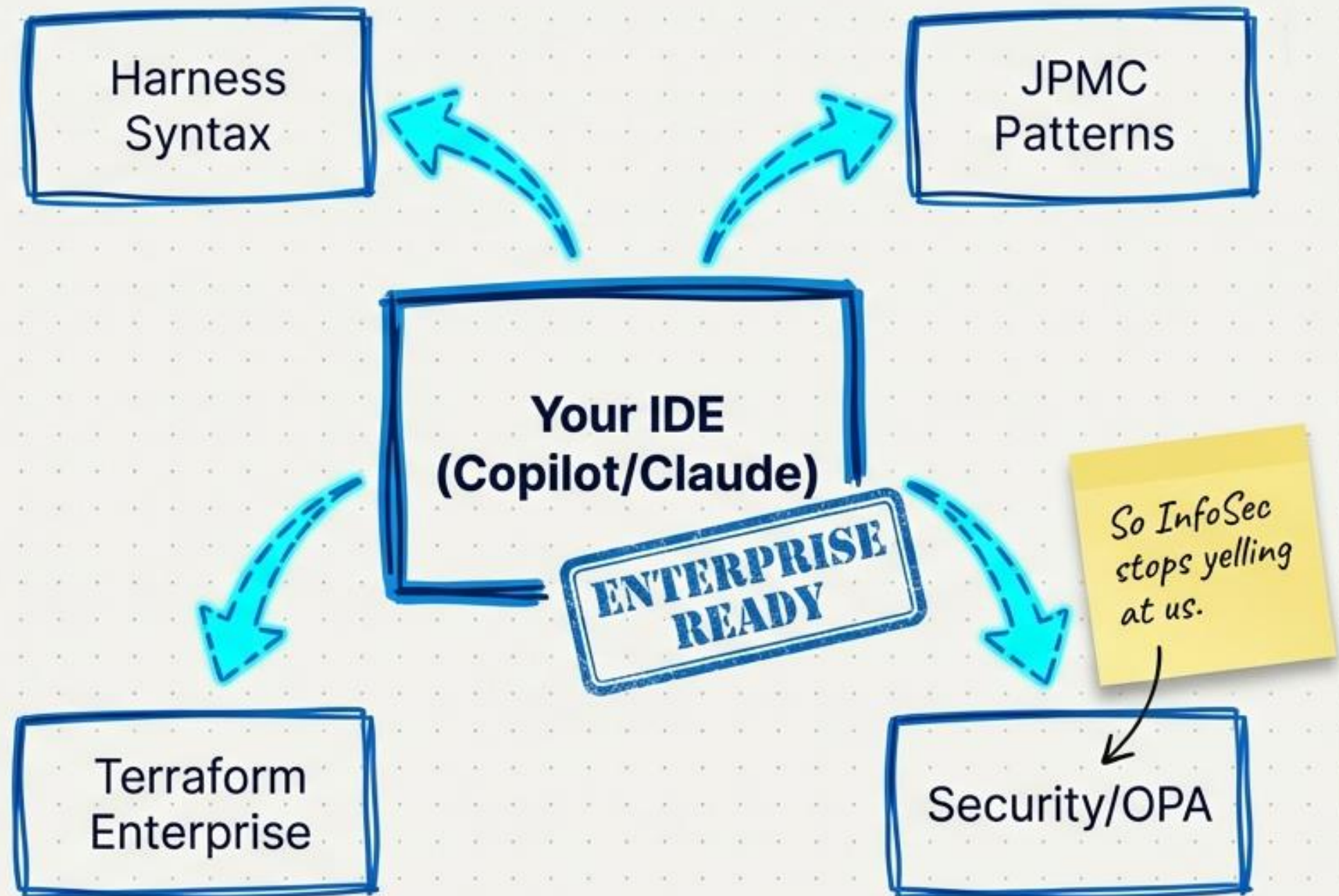
The SRE's Guide to Building, Deploying,
and Surviving Enterprise Pipelines

Plz don't
erase
- SRE Team

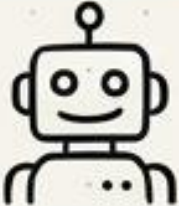
What exactly is this?

It's not just an autocomplete. It's an AI-assisted toolchain injected with deep Harness platform expertise and mandatory JPMC enterprise patterns.

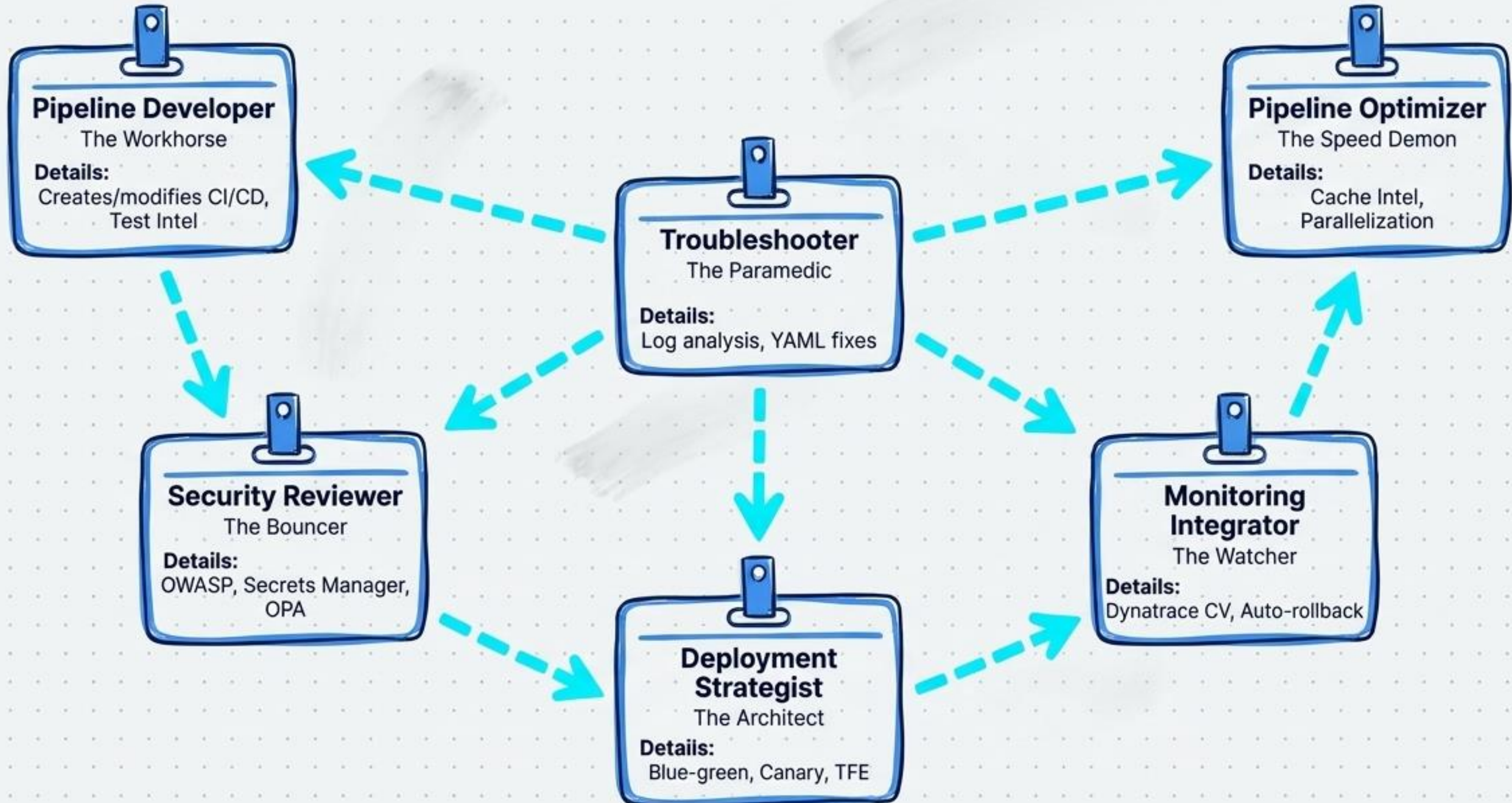
- ✓ Pre-built task templates
- ✓ Real-time YAML validation
- ✓ Automated JPMC compliance
- ✓ Built-in cost estimation



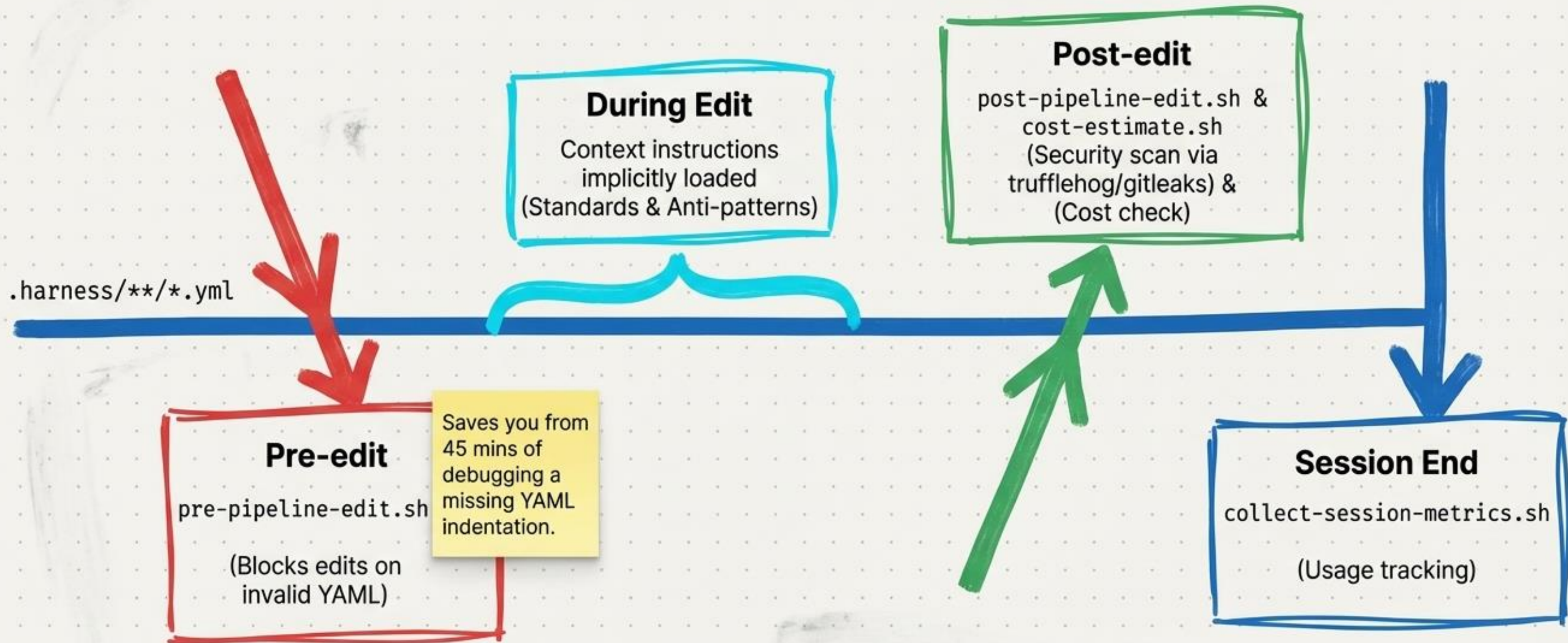
Choose Your Weapon: Agents vs. Prompts vs. Skills

 Agents ✓	 Prompts ✓	 Skills ✓
Trigger: <code>@Harness[Name]</code>	Trigger: Use prompt: [name]	Trigger: Auto-referenced / Read directly
Use When: Complex logic, multi-step reasoning, or debugging.	Use When: Step-by-step guidance for a well-defined task with placeholders.	Use When: You need reference docs or specific YAML patterns.
Example: Audit this pipeline for OWASP.	Example: Use prompt : create-ci-pipeline	Example: How does OPA work here?

Your New AI Teammates

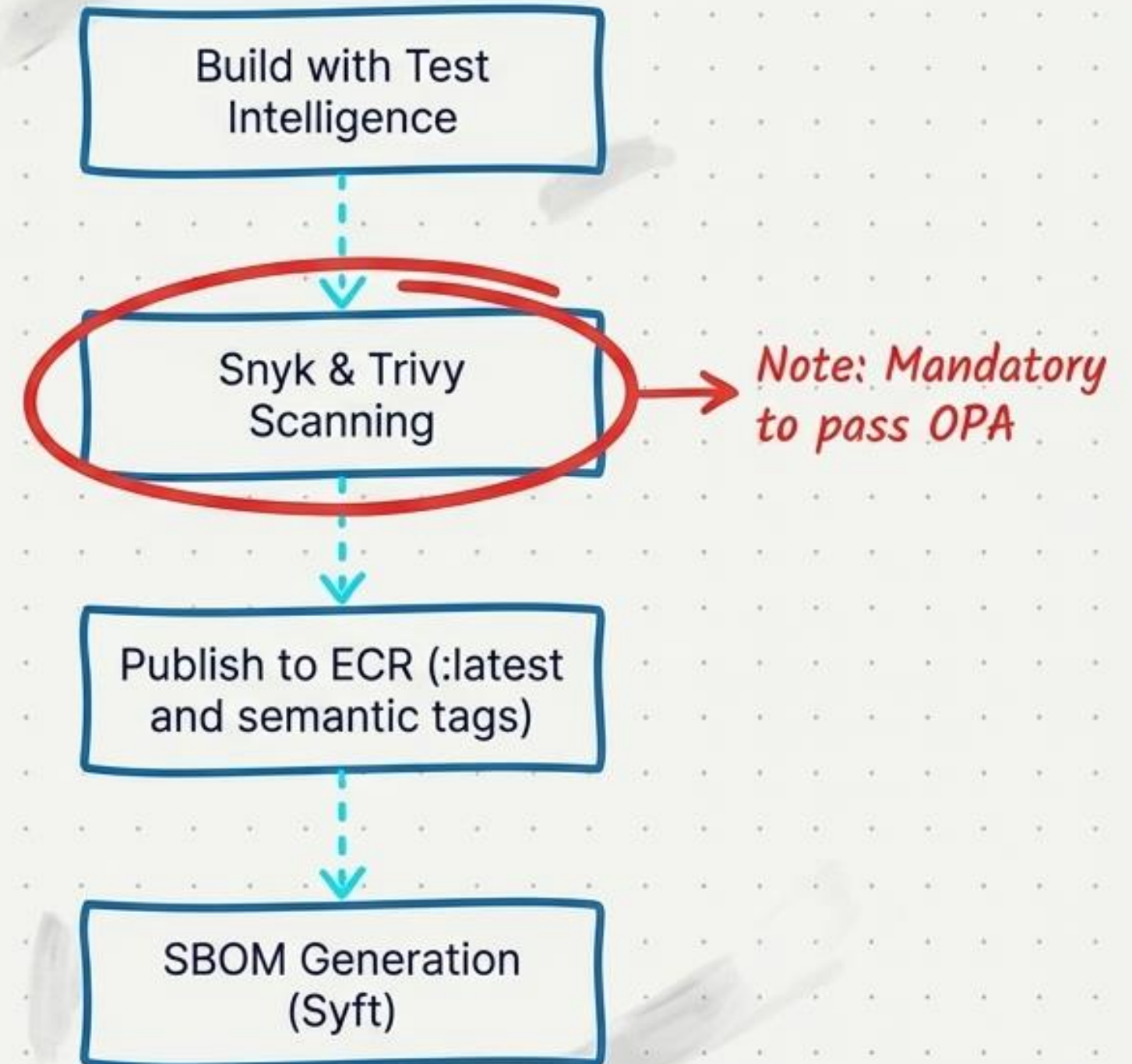


The Invisible Guardrails (Hook Architecture)

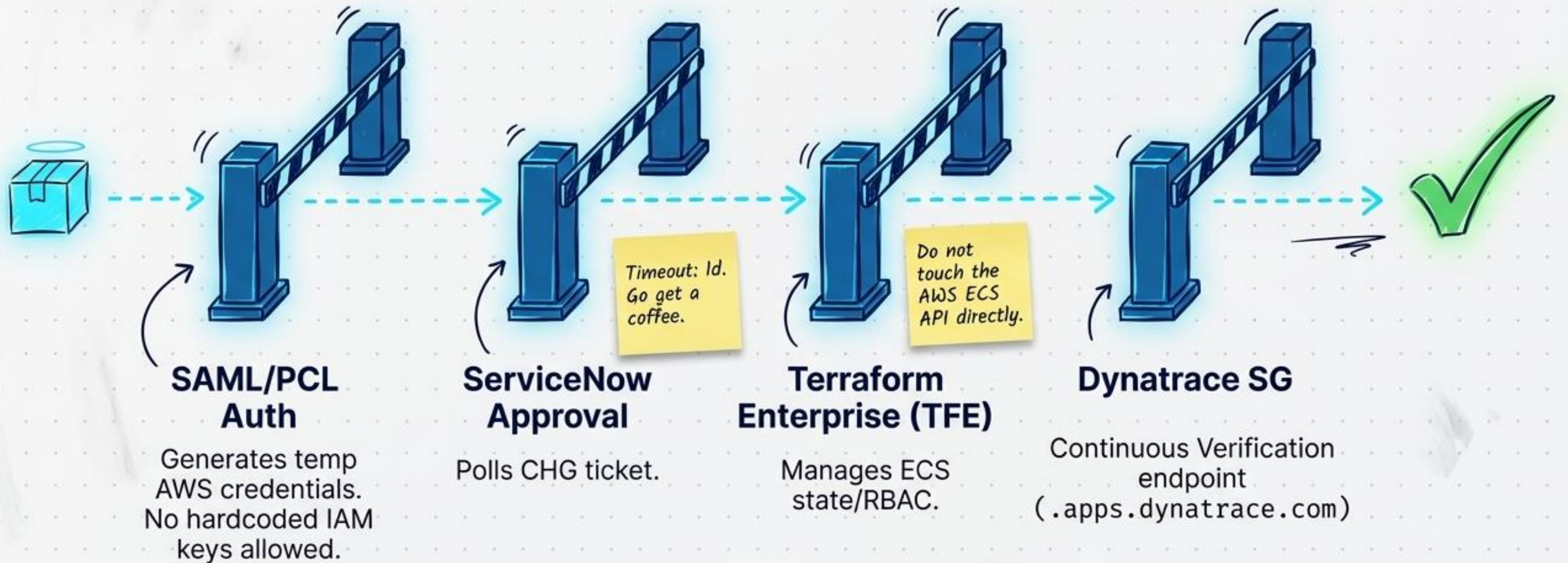


Phase 1: The Instant CI Pipeline

```
> Use prompt: create-ci-pipeline
Inputs:
- ${pipelineName}
- ${module}
- ${buildTool} (maven, npm, pip)
```



Phase 2: Surviving the JPMC Prod Deployment Gauntlet



Phase 3: Canary Deployments Without the Sweat

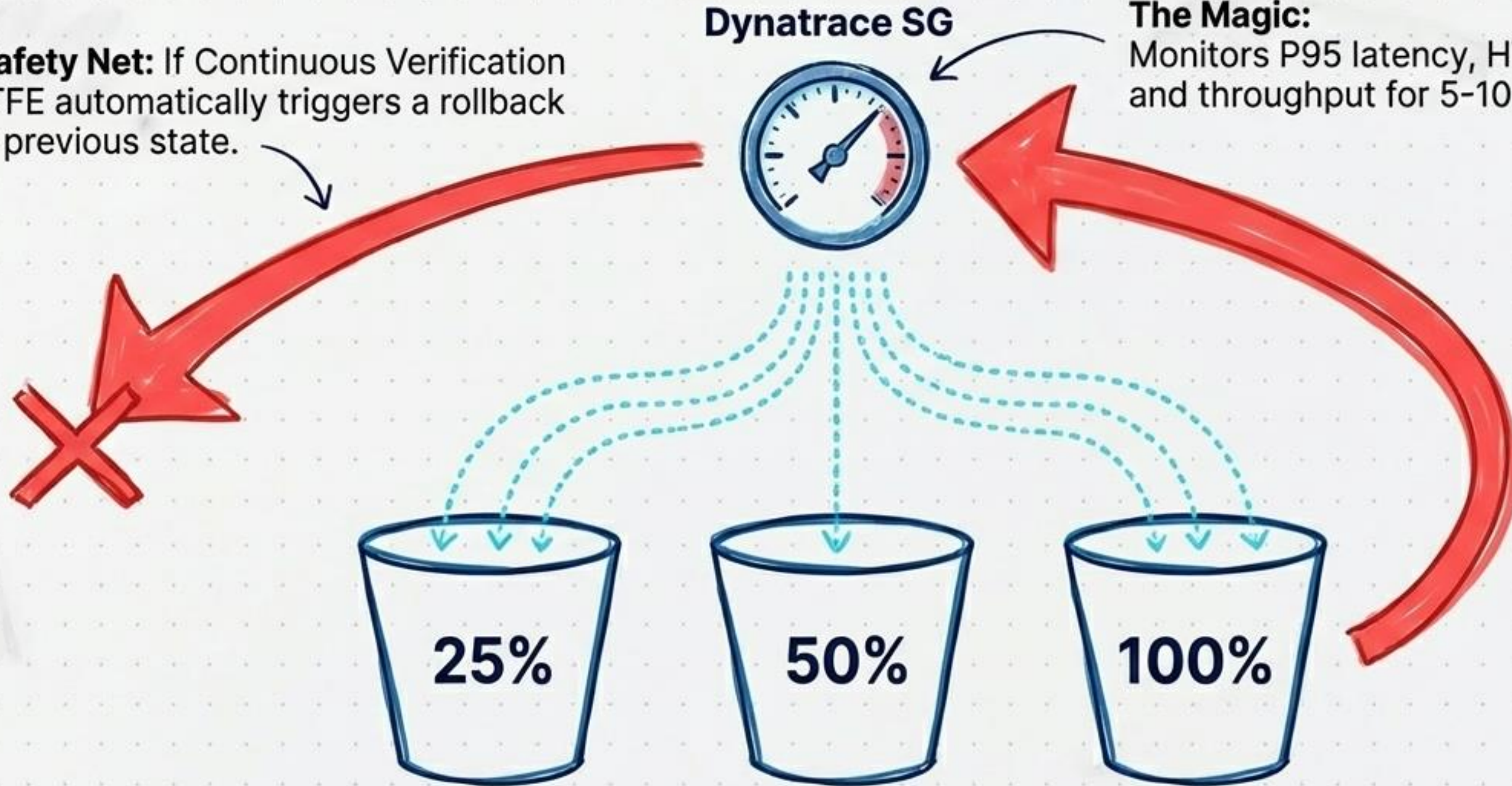
@HarnessDeploymentStrategist set up a canary rollout with CV.

The Safety Net: If Continuous Verification fails, TFE automatically triggers a rollback to the previous state.

Dynatrace SG

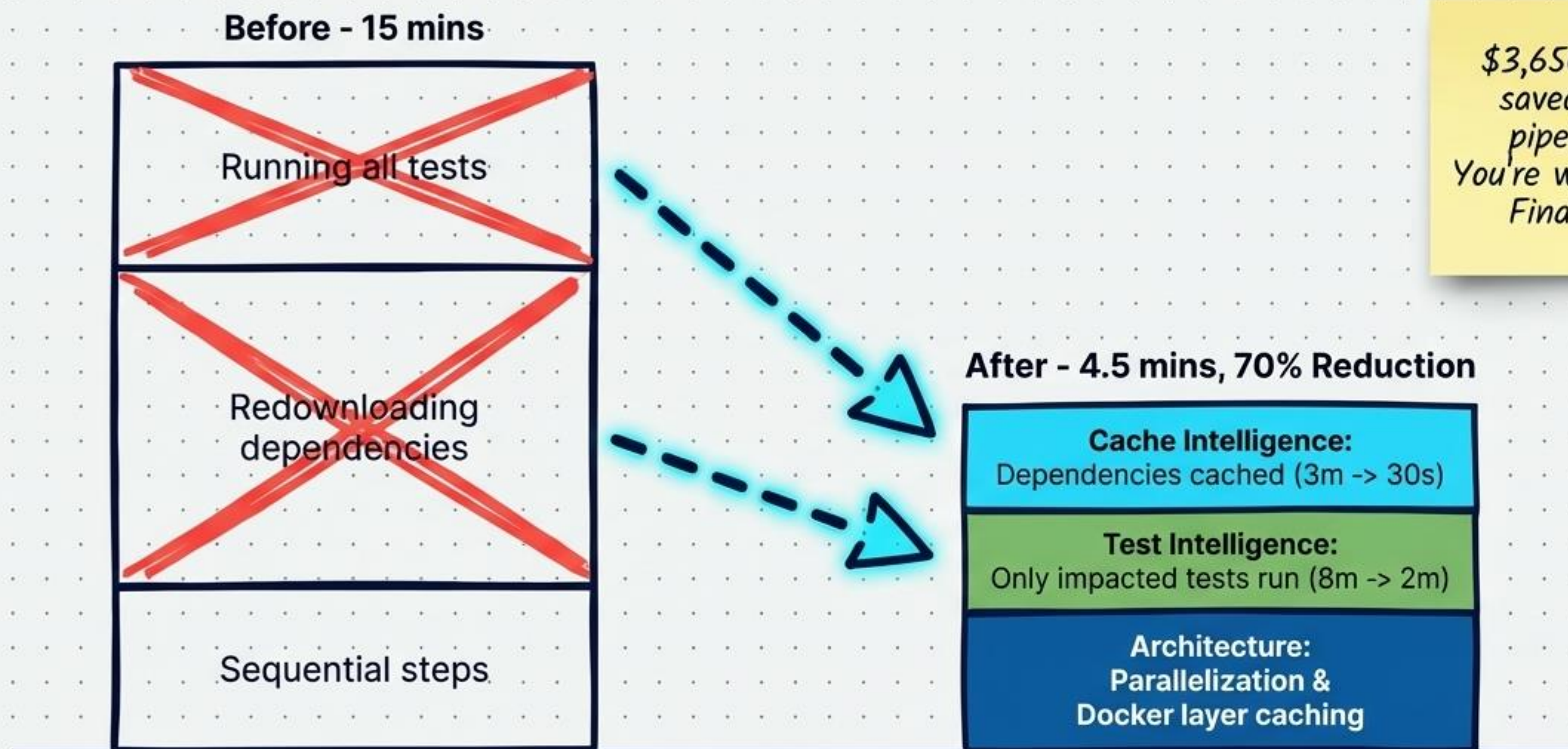
The Magic:

Monitors P95 latency, HTTP 5xx errors, and throughput for 5-10 minutes.



Phase 4: Slashing Build Times

@HarnessPipelineOptimizer optimize this 15m build.



Phase 5: The 2 AM Troubleshooter

@HarnessTroubleshooter analyze this failure log.



Error 1: OPA 'require-security-scan' evaluation failed?

Fix: Snyk/Trivy step must have type 'Security', not generic 'Run'.

Error 2: TFE workspace is locked?

Fix: Pending run in TFE or requires manual Force Unlock in TFE UI.

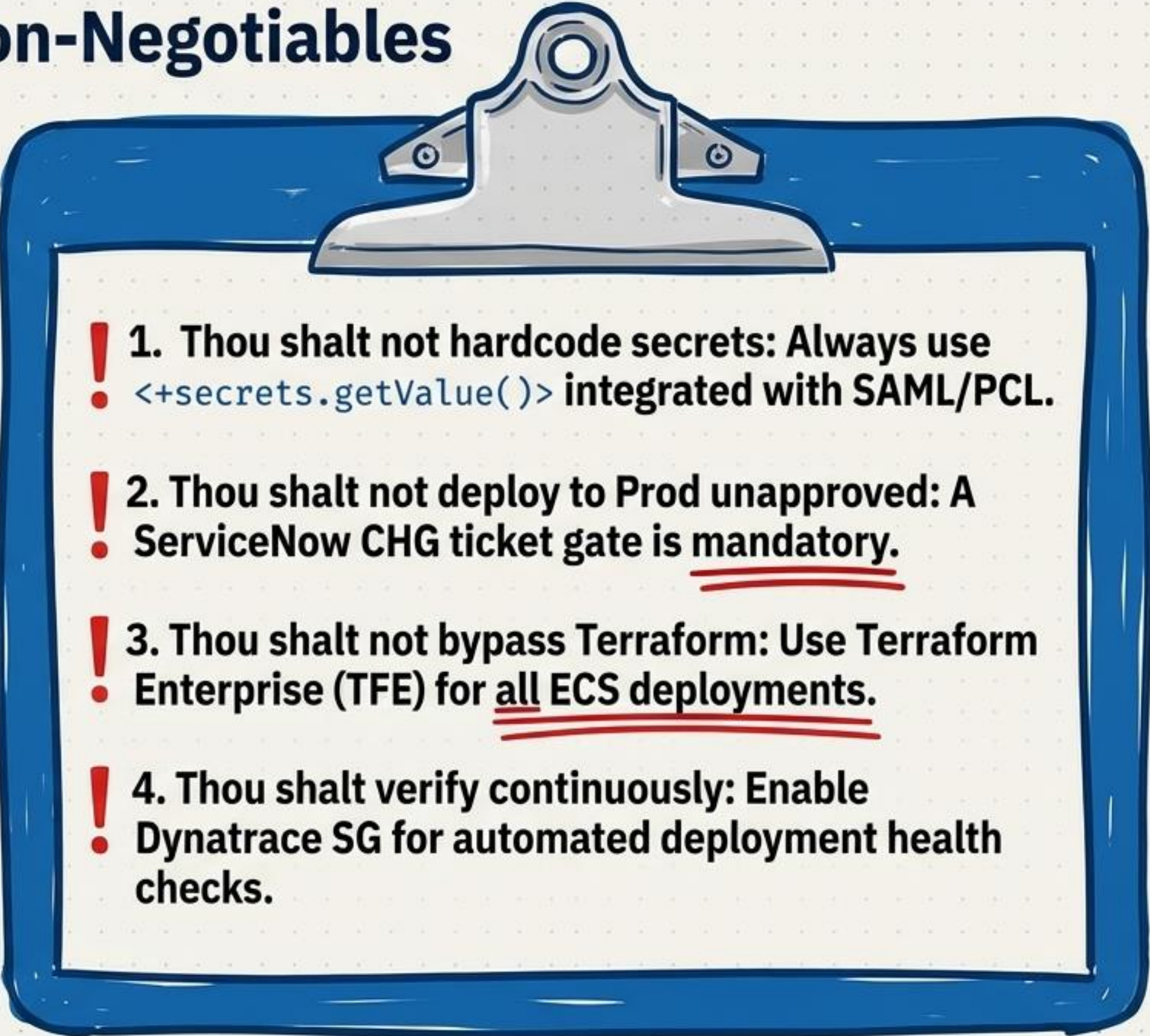
Error 3: Cache hit rate 0%?

Fix: Caching requires Harness Cloud infrastructure, not local delegates.

Pro-Tip: Always check the "Security" type for compliance steps.

Reminder: Force Unlock is a last resort!

The JPMC Non-Negotiables

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- ! 1. Thou shalt not hardcode secrets: Always use `<+secrets.getValue(>` integrated with SAML/PCL.
 - ! 2. Thou shalt not deploy to Prod unapproved: A ServiceNow CHG ticket gate is mandatory.
 - ! 3. Thou shalt not bypass Terraform: Use Terraform Enterprise (TFE) for all ECS deployments.
 - ! 4. Thou shalt verify continuously: Enable Dynatrace SG for automated deployment health checks.

Pro-Tip: Compliance isn't just a hurdle; it's your safety net. Adhere to these to avoid 2 AM wake-up calls!
- The SRE Team

Quick Start Cheat Sheet



Start a CI Pipeline:

> Use prompt: `create-ci-pipeline`

Deploy to Prod:

> `@HarnessPipelineDeveloper` create a CD pipeline for [Project]

Add a Canary Strategy:

> `@HarnessDeploymentStrategist` set up canary

Check Your Work:

> `@HarnessSecurityReviewer` audit this file

Fix a Failure:

> `@HarnessTroubleshooter` analyze this log

Cap your
markers.
Time to ship.

Where to look under the hood: `.github/agents/`,
`.github/skills/`, `.github/prompts/`